Rajinder Lal
Math 400

Math 400 Final Exam Part 5 Due by Wednesday, December 16 11:59 pm

Instructions: Please show an appropriate amount of work for each problem. Write legibly and clearly. Please circle your answer. Make sure to simplify all answers.

You can print out this exam and write your solutions directly on the exam. If you are unable to print out the exam, then you can write your solutions on separate pieces of paper. If you write your solutions on separate paper, then **make sure to put the problems in numerical order. Please write your name on the exam.**

You'll have to submit the exam to me in **Canvas**. Please scan your exam and upload your **single** PDF file to Canvas. **Look under Assignments – Exams – Final Exam Part 5.**

****This is an exam, and so you shouldn't be working on the exam with other people and/or use apps/websites that can solve the problem for you.**

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Name: Kanageshwaran Dhakshinamoorthy

Math 400 Final Exam Part 5 24 Total Points

(8 pts) 1. Find the derivative of the function using the definition of a derivative.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

$f(x) = \frac{10x}{x-2}$, $f(x+h) = \frac{10(x+h)}{(x+h)-2} = \frac{10x+10h}{x+h-2}$

$$f'(x) = \lim_{h\to 0} \frac{\left(\frac{10x+10h}{x+h-2}\right) - \left(\frac{10x}{x-2}\right)}{h}$$

$$f'(x) = \lim_{h\to 0} \frac{(10x+10h)(x-2) - 10x(x+h-2)}{h(x-2)(x+h-2)}$$

$$= \lim_{h\to 0} \frac{10x^2 + 10xh - 20x - 20h - 10x^2 - 10xh + 20x}{h(x^2 - 4x + hx - 2h + 4)}$$

$$= \lim_{h\to 0} \frac{-20h}{h(x^2 - 4x + hx - 2h + 4)}$$

$$= \lim_{h\to 0} \frac{-20}{(x^2 - 4x + hx + 2h + 4)} = -\frac{20}{x^2 - 4x + 4}$$

($hx \to 0$, $2h \to 0$)

$$\boxed{f'(x) = \frac{-20}{x^2 - 4x + 4}}$$

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(8 pts) 2. Evaluate the indefinite integral.

$$\int x^2\sqrt{x+4}\,dx$$

Let $u = x+4$, $du = dx$, $x = u-4$

$$= \int (u)^{1/2}(u-4)^2\,du$$

$$= \int (u^{1/2})(u^2-8u+16)\,du$$

$$= \int (u^{5/2} - 8u^{3/2} + 16u^{1/2})\,du$$

$$= \int u^{5/2}\,du - 8\int u^{3/2}\,du + 16\int u^{1/2}\,du$$

$$= \frac{u^{7/2}}{7/2} - 8\left(\frac{u^{5/2}}{5/2}\right) + 16\left(\frac{u^{3/2}}{3/2}\right) + C$$

$$= \left(\frac{2}{7}\right)u^{7/2} - \left(\frac{16}{5}\right)u^{5/2} + \left(\frac{32}{3}\right)u^{3/2} + C$$

$$= \frac{2}{7}(x+4)^{7/2} - \left(\frac{16}{5}\right)(x+4)^{5/2} + \left(\frac{32}{3}\right)(x+4)^{3/2} + C$$

$$\therefore \int x^2\sqrt{x+4}\,dx = \frac{2}{7}(x+4)^{7/2} - \frac{16}{5}(x+4)^{5/2} + \frac{32}{3}(x+4)^{3/2} + C$$

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(8 pts) 3. Evaluate the following integral. Please give an **exact answer. No decimal approximations.**

$$\int_0^{\pi} 3^{2+\cos x}(\sin x)\,dx$$

Let $u = 2+\cos x$, $du = -\sin x\,dx \Rightarrow -du = \sin x\,dx$

$$= \int_0^{\pi} 3^{u}(-du)$$

$$= \left[\frac{3^{u}}{\ln(3)}(-1) + C \right]_0^{\pi}$$

$$= \left[\frac{-3^{2+\cos x}}{\ln(3)} + C \right]_0^{\pi}$$

$$= \left[\frac{-3^{2+\cos(\pi)}}{\ln(3)}\right] + C - \left[-\frac{3^{2+\cos(0)}}{\ln(3)} + C\right]$$

$$= \frac{3^{2+\cos(0)}}{\ln(3)} - \frac{3^{2+\cos(\pi)}}{\ln(3)} = \frac{3^{2+\cos(0)} - 3^{2+\cos(\pi)}}{\ln(3)}$$

$$= \frac{3^{2+1} - 3^{2+(-1)}}{\ln(3)} = \frac{3^{3} - 3^{1}}{\ln(3)} = \frac{27-3}{\ln(3)}$$

$$= \frac{24}{\ln(3)}$$

$$\int_0^{\pi} 3^{2+\cos x}\sin(x)\,dx = \frac{24}{\ln(3)}$$

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